

NANYANG TECHNOLOGICAL UNIVERSITY

First Year Common Engineering Course

FE1073 Introduction to Engineering & Practices

Laboratory Manual
for
Experiment M2

CONSERVATION OF MOMENTUM

Laboratory : Mechanics of Materials (MAE)

Location : N3.2-B2-01

Session 2016/2017

1 INTRODUCTION

An attempt will be made here to demonstrate the principle of conservation of momentum in collisions of sliding masses on an almost frictionless air track.

On the air track, a slider moving at a particular constant velocity collides with another slider which is initially at rest. It will be shown experimentally that the total momentum of the colliding sliders is the same before and after collision. The total momentum is conserved only if there is no external force such as friction acting on the sliders. This explains why the experiment has to be carried out on an air track which exerts almost zero frictional force against the motion of the sliders.

Only one-dimensional collision in which the motion of the sliders is along a straight line is examined here. If a slider whose mass is m moves with a constant velocity $\mathbf{v}$, then its momentum is $m\mathbf{v}$. Note that $\mathbf{v}$ may be positive or negative depending on the direction of the slider. If the one-dimensional velocity of a slider in a particular direction is taken to be positive, then the velocity in the opposite direction is negative. Thus, the momentum $m\mathbf{v}$ of the slider may be positive or negative.

A collision between bodies is termed elastic or inelastic depending on whether there is a loss in the total kinetic energy of the bodies. If the total kinetic energy is conserved, the collision is elastic. Otherwise, the collision is said to be inelastic. In the case of an inelastic collision, the kinetic energy lost is converted into other forms such as heat energy or work done in deforming the bodies.

Consider now a slider of mass m_1 (slider 1) moving with constant velocity u_1 (along a straight line) before colliding with another slider of mass m_2 (slider 2) which is initially at rest. After the collision, the velocity of slider 1 is v_1 , while that of slider 2 is v_2 . If the total momentum of the sliders is conserved, we would expect

$$m_1 u_1 = m_1 v_1 + m_2 v_2 \quad \text{for one-dimensional collisions.} \quad (1)$$

An experimental verification of Eq. (1) will be carried out for two extreme types of collisions: (a) elastic (or nearly elastic) collisions in which sliders 1 and 2 separate with different velocities after colliding, and (b) completely inelastic collisions in which the two sliders stick to each other and move with the same velocity after colliding (that is, $v_1 = v_2$ for completely inelastic collisions).

Exercise 1. If the collision between sliders 1 and 2 is elastic then

$$\frac{1}{2} m_1 u_1^2 = \quad (2)$$

Exercise 2. If the two sliders have equal mass and if their collision is elastic, use Eqs. (1) and (2) to find the velocities of sliders 1 and 2, that is, to find v_1 and v_2 .

Exercise 3. For an elastic collision between two sliders, m_1 and m_2 , let u_1 and u_2 be the velocities before collision, v_1 and v_2 the velocities after collision. Determine the velocities v_i after collision in terms of the masses, m_i , and the velocities before collision, u_i .

Exercise 4. For a completely inelastic collision, that is, for the case in which $v_1 = v_2$, use Eq. (1) to show that the total kinetic energy of the sliders after the collision is definitely less than the total kinetic energy before the collision. (Find a formula for the total kinetic energy after the collision in terms of m_1 , m_2 and u_1 .)

Exercise 5. The coefficient of restitution, denoted by e , is defined as the ratio of final to initial relative velocity of the two sliders before and after the collision. Mathematically,

$$e = \frac{v_2 - v_1}{u_1 - u_2} \quad (3)$$

Determine the coefficient of restitution for elastic collision using the obtained results in Exercise 3 and completely inelastic collision.

2 OBJECTIVE

After you have finished this experiment, you should have a better understanding of:

- elastic and inelastic collisions,
- conservation of momentum in both elastic and inelastic collisions,
- kinetic energy of the system and its role in determining if a collision is elastic or inelastic, and
- the fact that the conservation of momentum is independent of the conservation of energy, that is, some kinetic energy is lost in inelastic collisions but the total momentum does not change (assuming that there is no external force involved).

3 OUTLINE OF EXPERIMENT

The tasks involved as described below.

- Sliders of the same mass are used to verify that equation (1) is true for an elastic collision independent of the velocity u_1 of slider 1 before the collision.
- Sliders having different masses are used to verify Eq. (1) for elastic collisions.
- An arrangement is made to ensure that the sliders move together as one after the collision in order to verify that Eq. (1) is true for inelastic collisions (in which $v_1 = v_2$).

4 APPARATUS AND SET-UP

Apparatus

- 1 air track, connected to air supply
- Sliders, with bumpers for elastic and inelastic collisions
- 2 digital photo-gate timers
- Ruler

Setting up

Set up the experiment following closely the instructions given in the guide sheet on your experiment bench. Switch on the air supply. Make sure that the air track is set up horizontally so that the sliders will remain stationary when they are placed on the track. If it is not, request the laboratory technician for assistance.

Position the sensors as shown in Figure 1. Switch on the digital photo timers.

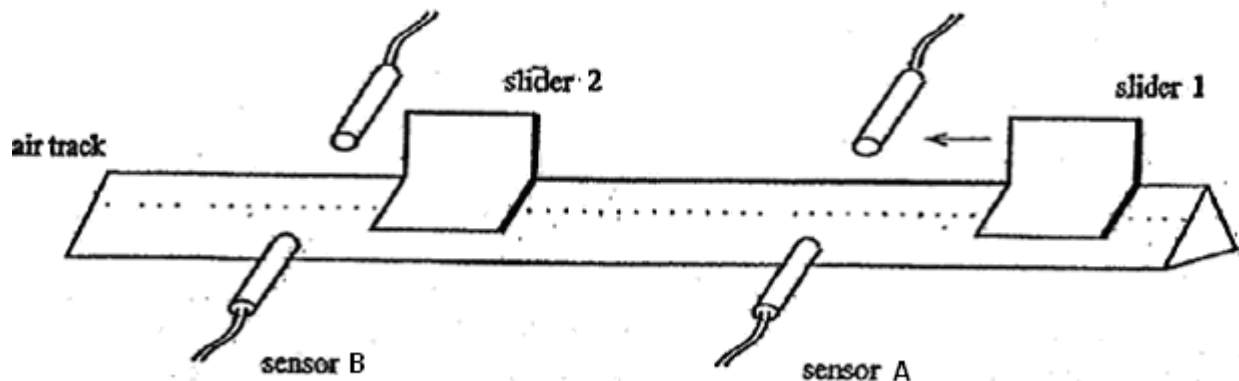


Figure 1. Experimental set-up.

5 EXPERIMENTAL PROCEDURE

To avoid making excessive noise, always switch off the air supply when the air track is not in use.

- Elastic collisions of sliders having the same mass.* Set up 2 sliders of (approximately) equal mass with spring bumpers for elastic collisions. Place slider 1 at the start of the track and slider 2 at rest between the sensors as shown in Figure 1. Push slider 1 to collide with slider 2 and record the readings t_A and $\hat{t}_B$ on the counters of sensors A and B respectively in Table 1. (Note that t_A is the elapsed time taken for slider 1 to pass sensor A before colliding with slider 2 and $\hat{t}_B$ is the elapsed time taken for slider 2 to pass sensor B after the collision.) Repeat the experiment two more times by giving slider

- 1 different initial velocities (before it collides with slider 2). Observe what happens to slider 1 after each collision. (Is your observation consistent with what you found out in **Exercise 2**?)
- b. *Elastic collisions of sliders having different masses.* Set up 2 sliders having different masses with spring bumpers for elastic collisions. Choose the mass of slider 2 to be about two times that of slider 1. Place slider 1 at the start of the track and slider 2 at rest between the sensors as shown in Figure 1. Push slider 1 to collide with slider 2 and record in Table 2 the elapsed time t_A of slider 1 before collision (as recorded by the counter of sensor A before the collision) and the elapsed times $\hat{t}_A$ and $\hat{t}_B$ of sliders 1 and 2 respectively after collision (as recorded by the counters of sensors A and B respectively). Note the direction of the motion of slider 1 after the collision and remember to take it into consideration when calculating the total momentum of the sliders. Repeat the experiment two more times using different initial velocities of slider 1.
- c. *Inelastic collisions of sliders having different masses.* Repeat the experiment described in (b) above using the bumpers for inelastic collision instead of the spring bumpers. The sliders should stick together after they collide. (Consult the laboratory technician if you do not know how this can be done.) Record in Table 4 the readings t_A and $\hat{t}_B$ on the counters of the sensors A and B respectively. Note there are two readings on the counter of the sensor B because two sliders pass through this sensor respectively. Determine the value of $\hat{t}_B$ and record it on the log sheet. Discuss with your supervisor the method you use for determining $\hat{t}_B$.

6 EVALUATION AND RESULTS

- a. *Elastic collisions of sliders having the same mass.* Using the experimentally obtained data recorded in Table 1, calculate the velocity of slider 1 before each of the collisions, that is, u_1 , and also the velocity of slider 2 after the collision, that is, v_2 . What can you say about the values of u_1 and v_2 obtained? What can you conclude about the total momentum as well as the total kinetic energy of the sliders before and after the collision?
- b. *Elastic collisions of sliders having different masses.* Using the experimentally obtained data recorded in Table 2, calculate the velocities of the sliders before and after each of the collisions. Complete Table 2. For each of the collisions recorded in Table 2, calculate the total momentum as well the total kinetic energy of the sliders before and after the collision and record your calculations in Table 3. What can you conclude from the results?
- c. *Inelastic collisions of sliders having different masses.* Using the experimentally obtained data recorded in Table 4, calculate the velocities of the sliders before and after each of the collisions. Complete Table 4. For each of the collisions recorded in Table 4, calculate

the total momentum as well the total kinetic energy of the sliders before and after the collision and record your calculations in Table 5. Is momentum conserved in each of the collision? What about the kinetic energy?

7 REPORT

7.1 Log Sheet and Formal Report

- a. What can you conclude from the results above?
- b. What are the possible sources of error in the experiment?
- c. From which Newton's law(s) of motion is the principle of conservation of momentum derived? Explain.

7.2 Formal Report Only

- a. Include your answers to Exercises 1-5 above.
- b. For the elastic collisions of sliders having different masses, use your results in the last three columns of Table 2 to calculate the coefficient of restitution associated with each case and the average coefficient of restitution. A theoretical prediction of the coefficient of restitution is derived in Exercise 5 above. Compare the experimental result with the theoretical one.
- c. For the inelastic collisions in the last part of experiment, use your results in the last column of Table 5 to calculate the average percentage of kinetic energy lost after a collision. A theoretical prediction of the percentage of the kinetic energy lost can also be derived using the formula for the kinetic energy after collision obtained as in Exercise 4 above. Compare the experimental result with the theoretical one.
- d. How would you rewrite equation (1) for two- or three-dimensional collisions?

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Experiment M2 : CONSERVATION OF LINEAR MOMENTUM

LOG SHEET

Name :

Date :

Group :

Submit to :

(a) *Elastic collisions of sliders having the same mass*

mass of slider 1 = kg mass of slider 2 = kg
length of slider 1 = m length of slider 2 = m

t_A (s)	$\hat{t}_B$ (s)	u_1 (m/s)	v_2 (m/s)

Table 1. Results for elastic collisions of sliders having the same mass.

Note. The data and results for a particular collision are to be entered into a specific row of the table, e.g. first row for the first collision.

(b) *Elastic collisions of sliders having different masses.*

mass of slider 1 = kg mass of slider 2 = kg
length of slider 1 = m length of slider 2 = m

t_A (s)	$\hat{t}_A$ (s)	$\hat{t}_B$ (s)	u_1 (m/s)	v_1 (m/s)	v_2 (m/s)

Table 2. Results for elastic collisions of sliders having different masses.

p (Ns)	$\hat{p}$ (Ns)	$\frac{p - \hat{p}}{p} \times 100\%$	E (Nm)	$\hat{E}$ (Nm)	$\frac{E - \hat{E}}{E} \times 100\%$

Table 3. Total momentum and kinetic energy before and after collision.

Notes. (1) The data and results for a particular collision are to be entered into a specific row of the tables above, e.g. first row for the first collision. (2) In calculating the velocities in Table 2, remember to take into consideration the directions of the sliders before and after the collisions. Velocities in opposite directions should have opposite signs. (3) In Table 3, p and $\hat{p}$ denote respectively the total momentum before and after collision, while E and $\hat{E}$ respectively the total kinetic energy before and after the collision.

(c) *Inelastic collisions of sliders having different masses*

mass of slider 1 = kg mass of slider 2 = kg
length of slider 1 = m length of slider 2 = m

t_A (s)	$\hat{t}_B$ (s)	u_1 (ms^{-1})	$v_1 = v_2$ (ms^{-1})

Table 4. Inelastic collisions of sliders having different mass.

p (Ns)	$\hat{p}$ (Ns)	$\frac{p - \hat{p}}{p} \times 100\%$	E (Nm)	$\hat{E}$ (Nm)	$\frac{E - \hat{E}}{E} \times 100\%$

Table 5. Total momentum and kinetic energy before and after collision.

Notes. (1) The data and results for a particular collision are to be entered into a specific row of the tables above, e.g. first row for the first collision. (2) In this experiment, sliders 1 and 2 have the same velocity $v_1 = v_2$ after the collision. (3) In Table 5, p and $\hat{p}$ denote respectively the total momentum before and after collision, while E and $\hat{E}$ respectively the total kinetic energy before and after the collision.

Discussion:

(a) For elastic collision of sliders having the same mass, what happened to slider 1 after each of the collisions?

(b) What can you conclude from your results concerning momentum transfer?

(c) What can you conclude from your results concerning energy transfer? If energy is lost, what happens to it?

(d) According to the coefficient of restitution defined in Eq. (3) and the data in Table 2, calculate the average coefficient of restitution? Is it reasonable?

(e) What are the possible sources of error for the experiment? Give estimates for the percentage errors involved.

(f) From which Newton's law(s) of motion is the principle of conservation of momentum derived? Explain.
